

TAIL AND DENSITY ASYMPTOTICS FOR GAUSSIAN EXPONENTIAL INTEGRALS

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ABSTRACT. We study upper tails of exponential integrals of continuous Gaussian processes (GPs) on a fixed compact interval with a continuous deterministic trend. We identify an optimised Jensen–Gaussian lower bound with an attained reproducing kernel Hilbert space variational problem. At a unique variance-maximising endpoint, this bound is asymptotically exact when the local squared-increment exponent exceeds the variance-loss exponent. Under matching local bounds, we also obtain second-order logarithmic asymptotics, including rougher regimes. Under continuity and nondegeneracy alone, we establish absolute continuity and a density-to-tail equivalent. We also give extensions to Gaussian random fields and to growing intervals for self-similar processes.

1. INTRODUCTION

Let $X(t), t \in \mathcal{A}_T, \mathcal{A}_T := [0, T], T > 0$ be a centred GP with almost surely continuous sample paths, covariance function R and variance function $v(t) := R(t, t), t \in \mathcal{A}_T$. Sample continuity and Gaussianity imply mean-square continuity and hence R is jointly continuous and v is continuous on $\mathcal{A}_T$. Assume hereafter that

$$(1) \quad v(t_0) > 0 \quad \text{for some } t_0 \in \mathcal{A}_T$$

and let $m : \mathcal{A}_T \rightarrow \mathbb{R}$ be a deterministic continuous trend. Given a measurable set $A \subset \mathcal{A}_T$ define

$$I_{X,m}(A) := \int_A e^{X(t)+m(t)} dt, \quad M_m(A) := \int_A e^{m(t)} dt.$$

Our principal object of study is the tail probability

$$p_T(b) := \mathbb{P}\{I_{X,m}(\mathcal{A}_T) > b\}, \quad b > 0.$$

There are numerous motivations for studying $p_T(b)$, see for example the applications mentioned in [1, 2, 3] and the references therein.

Precise tail asymptotics for smooth homogeneous Gaussian random fields were obtained in [4, 5], while density estimates and exact density asymptotics were later developed in [6]. Efficient simulation methods for exponential integrals, including Hölder continuous fields and conditional rare-event descriptions, were studied in [7, 2]. For related rare-event simulation and discretisation of Gaussian suprema, see also [8].

The recent contribution [3] discusses bounds and approximation for X being a fractional Brownian motion (fBM). Other recent related works include [9, 10, 11, 12]. A classical example of exact asymptotics for a non-stationary GP is the Brownian motion benchmark recalled below.

In a different asymptotic regime, [13] obtained sharp small-noise asymptotics for aggregated lognormal random fields, while [14] extended this approach to differentiable Gaussian-input functionals, with applications to random elliptic PDEs. These results concern shrinking excursions above the deterministic value as the noise amplitude tends to zero, rather than the fixed-noise limit $b \rightarrow \infty$ considered here.

Motivated by the recent findings of [3] for $A = [a^-, a^+] \subset \mathcal{A}_T$ with $a^- < a^+$, $\max_{t \in A} v(t) > 0$ and all $b > M_m(A)$, the optimised Jensen–Gaussian bound reads (see (69) below)

$$(2) \quad \mathbb{P}\{I_{X,m}(A) > b\} \geq \Psi(\rho_{A,m}(b)),$$

where Φ and $\varphi := \Phi'$ are the standard normal distribution function and density, respectively, and we set $\Psi := 1 - \Phi$. Here we define for $b > 0$

$$(3) \quad \rho_{A,m}(b) := \inf \left\{ \|h\|_{\mathcal{H}_X(A)} : h \in \mathcal{H}_X(A), \quad \int_A e^{m(t)+h(t)} dt \geq b \right\} \in [0, \infty),$$

where $\mathcal{H}_X(A)$ is the reproducing kernel Hilbert space (RKHS) of $X(t), t \in A$. Here $\rho_{A,m}(b) = 0$ for $b \leq M_m(A)$, while $\rho_{A,m}(b) > 0$ for $b > M_m(A)$.

In covariance-parametrised form this Gaussian energy is the optimisation cost in [13, Section 2] and in the functional setting of [14, Section 2]; their small-noise analysis uses restricted displacements. The corresponding Gibbs stationarity relation appears in [13, Section 2.2]. The Jensen–Gaussian bound underlying (2) and related moment estimates appear already in [3, Propositions 2.3 and 2.13(ii)].

Our Lemma A.1 establishes the global identity between the entropy-optimised Gaussian level and the full RKHS distance, together with attainment and the optimiser correspondence, under the standing continuity assumptions; degeneracy of the GP is allowed. Lemma A.2 derives the resulting probability and moment bounds.

Our analysis shows that the lower bound (2) is exact on the logarithmic scale, i.e.,

$$\log p_T(b) \sim -\frac{\rho_{A_T,m}(b)^2}{2}, \quad b \rightarrow \infty$$

see Lemma 2.1 below; in our notation $a_b \sim c_b$ means asymptotic equivalence as $b \rightarrow \infty$, i.e., $\lim_{b \rightarrow \infty} a_b/c_b = 1$. This leads to the following natural question.

Q1. *What is the precise asymptotic approximation of $p_T(b)$ as $b \rightarrow \infty$, and when is the optimised Gaussian lower bound (2) asymptotically exact?*

Two zero-trend examples illustrate the first question. For $X(t) = tZ$, $t \in [0, T]$, where Z is standard normal we have

$$(4) \quad \mathbb{P} \left\{ \int_0^T e^{tZ} dt > b \right\} = \Psi(u_T(b)), \quad b > 0,$$

where $u_T(b)$ is the unique solution of $\int_0^T e^{tu_T(b)} dt = b$. For $b > T$ we have $u_T(b) = \rho_{[0,T],0}(b)$ and hence the lower bound is an identity. For standard Brownian motion, however, the ratio of the tail to this lower bound tends to $\sqrt{2}$; see Example 3.1.

We answer Question 1 in the variance-dominated endpoint regime: when the local squared-increment exponent is strictly larger than the variance-loss exponent, the lower bound (2) is asymptotically exact. We also obtain a second-order logarithmic expansion in regimes where this separation fails.

The key to the exact asymptotics is the representation

$$(5) \quad \mathbb{P}\{I_{X,m}(A) > b\} = \mathbb{E}\{\Psi(\rho_{A,m}(b) + \varkappa(W))\}, \quad b > M_m(A),$$

valid on any sufficiently small compact interval $A \subset [0, T]$ of positive length containing t_0 with $v(t_0) > 0$. Here W is the residual Gaussian process obtained by regression along an optimal RKHS direction, and $\varkappa : C(A) \rightarrow \mathbb{R}$ is continuous with $\varkappa(W) \leq 0$ almost surely; both may depend on b . See Lemma A.4 and Remark A.5 below.

Density existence for $I_{X,m}(\mathcal{A}_T)$ under additional regularity assumptions is established in [6, Theorem 1]. The Gaussian-quantile argument in [7] also implies a density-to-tail equivalent under Hölder regularity assumptions. This motivates the following question under our standing assumptions.

Q2. *Under continuity and nondegeneracy alone, does $I_{X,m}(\mathcal{A}_T)$ admit a density whose upper-tail behaviour can be expressed in terms of $p_T(b)$?*

We establish the density-to-tail equivalent under these assumptions, allowing the variance to vanish at some locations. Combining it with the endpoint tail equivalent gives an exact density approximation in terms of the variational level.

Beyond the exact endpoint regime, bounds and logarithmic asymptotics remain useful. The leading logarithmic rate holds under the standing assumptions, while its second-order refinement uses local Hölder-type conditions near the dominating variance point.

In the non-stationary settings of the paper it is natural to consider also growing domains, i.e., $T = T_b$ grows with b , which leads to our last question:

Q3. *What is the asymptotic behaviour of $p_{T_b}(b)$ when $b \rightarrow \infty$ and $T_b \rightarrow \infty$?*

We shall address the above question for self-similar GP's distinguishing the growth conditions needed for logarithmic and exact asymptotics.

Our contributions are as follows.

- (a) We identify the entropy-optimised Gaussian level with an attained RKHS variational problem and derive non-asymptotic lower and upper bounds for fixed b . Under continuity and nondegeneracy alone, these bounds also yield the leading logarithmic asymptotics of $p_T(b)$.
- (b) At a unique variance-maximising endpoint, we obtain a second-order logarithmic expansion under weak assumptions, which are satisfied for instance for fBM with $H < 1/2$.
- (c) When the local squared-increment exponent is strictly larger than the variance-loss exponent, we prove that the lower bound (2) is asymptotically exact. Besides fBM with $H > 1/2$, this covers further stationary-increment and Volterra-type Gaussian processes.
- (d) Under continuity and nondegeneracy alone, we prove absolute continuity and a density-to-tail equivalent; in the exact endpoint regime this gives an explicit Gaussian-density equivalent.
- (e) We derive leading, second-order, and exact asymptotics on growing domains for self-similar Gaussian processes. We also indicate how the exact tail and density results extend to Gaussian fields on compact rectangles.

Organisation of the paper:

Section 2 presents the bounds and logarithmic asymptotics, followed by the exact tail and density asymptotics and growing-domain results for self-similar processes. Section 3 gives examples and boundary cases, including a fractional Brownian sheet covered by Remark 2.7. The proofs are in Section 4. Appendix A collects the variational and Gaussian tools.

2. MAIN RESULTS

Consider X, m, T , and $\mathcal{A}_T$ as in the Introduction and put

$$\sigma_*^2 := \max_{t \in \mathcal{A}_T} v(t) \in (0, \infty).$$

We shall derive next a general upper bound and logarithmic asymptotics for $p_T(b)$ under the standing assumptions. We then refine the logarithmic asymptotics when v attains its unique maximum at T .

The leading logarithmic tail rate was established under Hölder regularity assumptions in [7]. The following preliminary result derives this rate under the standing continuity assumptions and relates it to the variational level.

Throughout the fixed-domain results put

$$L_b := \log \frac{b}{M_m(\mathcal{A}_T)}, \quad b > M_m(\mathcal{A}_T) := \int_{\mathcal{A}_T} e^{m(t)} dt.$$

The following preliminary lemma recovers the leading logarithmic rate of [7, Lemma 4.3] under weaker regularity assumptions and relates it to the variational level. Its proof requires only continuity and nondegeneracy at one point.

Lemma 2.1. *For all $b \geq M_m(\mathcal{A}_T)e^{\sigma_*^2}$ we have*

$$(6) \quad p_T(b) \leq \exp \left\{ -\frac{L_b^2}{2\sigma_*^2} \right\}$$

and moreover as $b \rightarrow \infty$

$$(7) \quad \log \mathbb{P}\{I_{X,m}(\mathcal{A}_T) > b\} \sim -\frac{\rho_{\mathcal{A}_T,m}(b)^2}{2} \sim -\frac{L_b^2}{2\sigma_*^2} \sim -\frac{(\log b)^2}{2\sigma_*^2}.$$

In order to refine the logarithmic asymptotics in (7), suppose next that T is the unique maximiser of the variance function v on $\mathcal{A}_T$ and that, for some

$$\alpha \in (0, 2], \quad \beta > 0, \quad \varepsilon \in (0, T],$$

the following local upper bounds

$$(8) \quad \mathbb{E}\{(X(s) - X(t))^2\} \leq C_d |s - t|^\alpha, \quad s, t \in [T - \varepsilon, T]$$

and

$$(9) \quad 0 \leq \sigma_*^2 - v(t) \leq C_v (T - t)^\beta, \quad t \in [T - \varepsilon, T]$$

are valid for some positive finite constants C_d, C_v .

We shall also assume that at least one of the following matching lower bounds holds for the same exponents after decreasing ε if necessary, namely

$$(10) \quad \alpha \leq \beta, \quad \mathbb{E}\{(X(s) - X(t))^2\} \geq c_d |s - t|^\alpha, \quad s, t \in [T - \varepsilon, T],$$

$$(11) \quad \beta \leq \alpha, \quad \sigma_*^2 - v(t) \geq c_v (T - t)^\beta, \quad t \in [T - \varepsilon, T],$$

for some positive finite constants c_d, c_v . Consequently, when $\alpha = \beta$ either matching lower bound is sufficient.

Proposition 2.2. *Under the preceding assumptions, with $\gamma := \min\{\alpha, \beta\}$ we have*

$$(12) \quad \log \mathbb{P}\{I_{X,m}(\mathcal{A}_T) > b\} = -\frac{L_b^2}{2\sigma_*^2} - \frac{L_b \log L_b}{\gamma \sigma_*^2} + O(L_b), \quad b \rightarrow \infty.$$

The lower estimate in (12) uses only the local upper bounds (8)–(9); the matching lower bounds are used for the upper estimate.

Corollary 2.3. *Under the assumptions and notation of Proposition 2.2 we have as $b \rightarrow \infty$*

$$(13) \quad \rho_{\mathcal{A}_T,m}(b)^2 = \frac{L_b^2}{\sigma_*^2} + \frac{2}{\gamma \sigma_*^2} L_b \log L_b + O(L_b).$$

Next, in addition to the standing assumptions suppose that T is the unique maximiser of v on $\mathcal{A}_T$ and that the following endpoint conditions hold for some $\varepsilon \in (0, T]$.

(C1) There are constants $0 < c_v \leq C_v < \infty$ and an exponent $\beta > 0$ such that

$$(14) \quad c_v (T - t)^\beta \leq \sigma_*^2 - v(t) \leq C_v (T - t)^\beta, \quad t \in [T - \varepsilon, T].$$

(C2) There are an exponent $\alpha \in (\beta, 2]$ and a constant $C_d > 0$ such that (8) holds on $[T - \varepsilon, T]$.

Theorem 2.4. *If T is the unique maximiser of v and Assumptions (C1) and (C2) hold, then*

$$(15) \quad \mathbb{P}\{I_{X,m}(\mathcal{A}_T) > b\} \sim \Psi(\rho_{\mathcal{A}_T,m}(b)), \quad b \rightarrow \infty$$

and

$$(16) \quad \rho_{\mathcal{A}_T,m}(b)^2 = \frac{L_b^2}{\sigma_*^2} + \frac{2}{\beta \sigma_*^2} L_b \log L_b + O(L_b).$$

The inequality $\alpha > \beta$ in Assumption (C2) describes the variance-dominated endpoint regime. The example $X(t) = tZ, t \geq 0$ in (4) satisfies these assumptions with $\beta = 1$ and $\alpha = 2$ and has an exact Gaussian-tail identity. The Brownian case in Example 3.1 shows that the coefficient 1 need not persist when the exponents are equal.

Corollary 2.5. *Suppose that $v(t) < v(T) = \sigma_*^2$ for every $t \in [0, T)$, that v is differentiable at T from within $\mathcal{A}_T$ and that*

$$(17) \quad v'(T) > 0.$$

If Assumption (C2) holds with $\beta = 1$, then the conclusion of Theorem 2.4 is valid.

Remark 2.6. The exponents in Assumptions (C1) and (C2) necessarily satisfy

$$(18) \quad \alpha \leq 2\beta.$$

Indeed, for $0 < h \leq \varepsilon$, Cauchy–Schwarz and the endpoint assumptions give

$$c_v h^\beta \leq \sigma_*^2 - v(T - h) \leq 2\sigma_* (\mathbb{E}\{(X(T) - X(T - h))^2\})^{1/2} \leq 2\sigma_* \sqrt{C_d} h^{\alpha/2}.$$

Letting $h \downarrow 0$ proves (18). Hence the admissible exponents satisfy

$$\beta < \alpha \leq \min\{2, 2\beta\}.$$

The remainder $O(L_b)$ in (16) is in general optimal; see Example 3.2.

Remark 2.7. The exact endpoint result extends to a centred Gaussian random field $X(t), t \in \mathcal{A}_T := [0, T]^d$, with continuous sample paths, $m \in C(\mathcal{A}_T)$ and $d \geq 2$. Define $I_{X,m}(\mathcal{A}_T)$, $M_m(\mathcal{A}_T)$ and $\rho_{\mathcal{A}_T,m}(b)$ by the same formulas, using d -dimensional Lebesgue measure and the RKHS of X on $\mathcal{A}_T$. Suppose that $\mathbf{T} := (T, \dots, T)$ is the unique maximiser of v on $\mathcal{A}_T$ with $v(\mathbf{T}) =: \sigma_*^2 > 0$. For some $\varepsilon \in (0, T]$, put $U := [T - \varepsilon, T]^d$ and assume that, for positive constants c_v, C_v, C_d ,

$$(19) \quad c_v \sum_{i=1}^d (T - t_i)^{\beta_i} \leq \sigma_*^2 - v(t) \leq C_v \sum_{i=1}^d (T - t_i)^{\beta_i}, \quad t \in U,$$

$$\mathbb{E}\{(X(s) - X(t))^2\} \leq C_d \sum_{i=1}^d |s_i - t_i|^{\alpha_i}, \quad s, t \in U,$$

where $0 < \beta_i < \alpha_i \leq 2$ for every i . Writing $L_b := \log\{b/M_m(\mathcal{A}_T)\}$ and $\gamma_\beta := \sum_{i=1}^d \beta_i^{-1}$, we obtain (15) and

$$(20) \quad \rho_{\mathcal{A}_T,m}(b)^2 = \frac{L_b^2}{\sigma_*^2} + \frac{2\gamma_\beta}{\sigma_*^2} L_b \log L_b + O(L_b), \quad b \rightarrow \infty.$$

The proof follows the same localisation and Gaussian regression argument, using rectangles with side lengths $r_{i,b} = \{(\log L_b)^2 / L_b\}^{1/\beta_i}$. Indeed, $\lim_{b \rightarrow \infty} L_b \sum_i r_{i,b}^{\alpha_i} = 0$ and (19) implies a polynomial covering bound after rescaling to $[0, 1]^d$, so the residual Gaussian estimates remain valid. For (20), product Laplace bounds and testing with the uniform density on $\prod_i [T - L_b^{-1/\beta_i}, T]$ replace the one-dimensional estimates.

2.1. Density asymptotics. We next answer Question 2. Put $I := I_{X,m}(\mathcal{A}_T)$ and $Y := \log I$, and denote the distribution function of Y by F_Y . The density-to-tail relation below requires only the standing assumptions from the Introduction. The endpoint assumptions yield the subsequent Gaussian-density corollary.

Theorem 2.8. *Under the standing assumptions, I and Y are absolutely continuous, $F'_{Y,+}(B)$ exists for all sufficiently large B , and there is a version f_I of the density of I such that*

$$(21) \quad f_I(b) = \frac{1}{b} F'_{Y,+}(\log b)$$

for all sufficiently large b . As $b \rightarrow \infty$

$$(22) \quad f_I(b) \sim \frac{\log b}{\sigma_*^2 b} \mathbb{P}\{I > b\} \sim \frac{\rho_{\mathcal{A}_T,m}(b)}{\sigma_* b} \mathbb{P}\{I > b\}.$$

Corollary 2.9. *If T is the unique maximiser of the variance function v and Assumptions (C1) and (C2) hold, then the density version in Theorem 2.8 satisfies*

$$(23) \quad f_I(b) \sim \frac{\varphi(\rho_{\mathcal{A}_T,m}(b))}{\sigma_* b}, \quad b \rightarrow \infty.$$

For smooth homogeneous Gaussian fields satisfying the hypotheses of [6, Theorem 3], combining their density asymptotic with the corresponding tail asymptotic yields, with f_Y denoting the log-integral density and σ^2 the common field variance

$$f_Y(B) \sim \frac{B}{\sigma^2} \mathbb{P}\{Y > B\}, \quad B \rightarrow \infty.$$

In the field setting of Remark 2.7, Lemma A.7 gives absolute continuity, and the proof of Theorem 2.8 applies on $C(\mathcal{A}_T)$. Thus (22) and (23) hold as well, with the density version chosen as in Theorem 2.8.

2.2. Growing domains for self-similar processes. We assume next that $m \equiv 0$ and that $X(t), t \geq 0$ is a centred Gaussian process with almost surely continuous sample paths which is κ -self-similar for some $\kappa > 0$, i.e.,

$$\{X(Tt), t \in [0, 1]\} \stackrel{d}{=} \{T^\kappa X(t), t \in [0, 1]\}, \quad T > 0.$$

Set $\sigma_1^2 := \text{Var}(X(1)) > 0$ and write

$$p_{T_b}(b) := \mathbb{P} \left\{ \int_0^{T_b} e^{X(t)} dt > b \right\}.$$

For the refinements below consider the local increment bound

$$(24) \quad \mathbb{E}\{(X(s) - X(t))^2\} \leq C_d |s - t|^\alpha, \quad s, t \in [1 - \varepsilon, 1],$$

where $\varepsilon \in (0, 1]$, $\alpha \in (0, 2]$ and $C_d > 0$. When $\alpha < 1$ we shall also use the matching lower bound

$$(25) \quad \mathbb{E}\{(X(s) - X(t))^2\} \geq c_d |s - t|^\alpha, \quad s, t \in [1 - \varepsilon, 1]$$

for some $c_d > 0$.

Proposition 2.10. *Let $T_b \rightarrow \infty$ and $L_b := \log(b/T_b) \rightarrow \infty$ as $b \rightarrow \infty$.*

(i) *If $T_b^\kappa = o(L_b)$, then*

$$(26) \quad \log p_{T_b}(b) \sim -\frac{\rho_{\mathcal{A}_{T_b},0}(b)^2}{2} \sim -\frac{L_b^2}{2\sigma_1^2 T_b^{2\kappa}} \sim -\frac{(\log b)^2}{2\sigma_1^2 T_b^{2\kappa}}.$$

(ii) *If $T_b^{2\kappa} = o(L_b)$ and (24) holds, together with (25) when $\alpha < 1$, then*

$$(27) \quad \log p_{T_b}(b) = -\frac{L_b^2}{2\sigma_1^2 T_b^{2\kappa}} - \frac{L_b \log L_b}{\gamma \sigma_1^2 T_b^{2\kappa}} + O\left(\frac{L_b}{T_b^{2\kappa}} + \log L_b\right),$$

where $\gamma := \min\{\alpha, 1\}$.

(iii) *If (24) holds with $\alpha \in (1, 2]$ and*

$$(28) \quad T_b^{2\kappa} = O(L_b),$$

then

$$(29) \quad p_{T_b}(b) \sim \Psi(\rho_{\mathcal{A}_{T_b},0}(b))$$

and

$$(30) \quad \rho_{\mathcal{A}_{T_b},0}(b)^2 = \frac{1}{T_b^{2\kappa}} \left\{ \frac{L_b^2}{\sigma_1^2} + \frac{2L_b \log L_b}{\sigma_1^2} + O(L_b) \right\}.$$

3. EXAMPLES

Example 3.1. Let $X = B$ be standard Brownian motion and $m = 0$. The law of $I_{X,0}([0, T])$ is known through Yor's representation involving the Hartman–Watson density; see [15], [1], and [16]. Write $A_t^{(0)} := \int_0^t e^{2B(s)} ds$. Brownian scaling gives $I_{X,0}([0, T]) \stackrel{d}{=} 4A_{T/4}^{(0)}$. For $b > T$, let $w_b > 0$ be the unique solution of

$$\frac{\sinh(w_b)}{w_b} = \frac{b}{T}.$$

If f_T denotes the density of $I_{X,0}([0, T])$, then [17, Theorem 2.1] with this scaling yield

$$(31) \quad f_T(b) \sim \frac{1}{b\sqrt{\pi T}} \exp\left\{-\frac{w_b - w_b^2/2}{T}\right\}, \quad b \rightarrow \infty.$$

Standard Brownian motion has variance-loss and squared-increment exponents both equal to 1, so the strict scale separation in Theorem 2.4 fails. In the zero-trend case, the explicit solution of the Brownian variational problem in [18, Proposition 12], after time rescaling and using the active constraint in Lemma A.1, gives

$$(32) \quad \rho_{[0,T],0}(b)^2 = \frac{w_b^2 - 2w_b \tanh(w_b/2)}{T}.$$

The Brownian density f_T is continuous, as also follows from [17, Eq. (6)] and scaling. Hence for all sufficiently large b it agrees with the density version in Theorem 2.8. That theorem gives

$$\mathbb{P}\{I_{X,0}([0, T]) > b\} \sim \frac{Tb}{\log(b/T)} f_T(b).$$

Since $w_b \sim \log(b/T)$, $\rho_{[0,T],0}(b) \sim w_b/\sqrt{T}$, and $w_b\{1 - \tanh(w_b/2)\} \rightarrow 0$ as $b \rightarrow \infty$, combining (31)–(32) with Mills' formula gives

$$(33) \quad \mathbb{P}\{I_{X,0}([0, T]) > b\} \sim \sqrt{2} \Psi(\rho_{[0,T],0}(b)) \sim \frac{Tb}{\log(b/T)} f_T(b), \quad b \rightarrow \infty.$$

Example 3.2. The remainder $O(L_b)$ in (16) cannot in general be replaced by $o(L_b)$. Indeed, take $X(t) = t^2 Z$ on $[0, T]$, where Z is standard normal, and $m = 0$. Then $\sigma_*^2 = T^4$, and Assumptions (C1) and (C2) hold with $\beta = 1$ and $\alpha = 2$. The RKHS consists of $h_a(t) = at^2$, with $\|h_a\|_{\mathcal{H}_X} = |a|$. Since $a \mapsto \int_0^T e^{at^2} dt$ is strictly increasing, for $b > T$ its unique solution a_b at level b is positive and equals $\rho_{[0,T],0}(b)$. Put $z_b := T^2 a_b$ and $L := \log(b/T)$. Substitution $x = z_b(1 - u)$ and dominated convergence, using $e^{-2x+x^2/z_b} \leq e^{-x}$ for $0 \leq x \leq z_b$ imply

$$\frac{b}{T} = \int_0^1 e^{z_b u^2} du \sim \frac{e^{z_b}}{2z_b}, \quad z_b = L + \log L + \log 2 + o(1).$$

Consequently, we obtain

$$\rho_{[0,T],0}(b)^2 = \frac{L^2 + 2L \log L + 2(\log 2)L}{T^4} + o(L),$$

which proves the assertion.

Example 3.3. Setting $X = B_H$ with $H \in (0, 1)$ and

$$\gamma_H := \min\{2H, 1\}, \quad L_b := \log\{b/M_m([0, T])\}$$

we have

$$(34) \quad \log \mathbb{P}\left\{\int_0^T e^{B_H(t)+m(t)} dt > b\right\} = -\frac{L_b^2}{2T^{2H}} - \frac{L_b \log L_b}{\gamma_H T^{2H}} + O(L_b).$$

For fractional Brownian motion

$$v(t) = t^{2H}, \quad d_{B_H}(s, t)^2 = |s - t|^{2H},$$

and

$$T^{2H} - t^{2H} \sim 2HT^{2H-1}(T - t), \quad t \uparrow T.$$

Proposition 2.2 therefore applies with

$$\alpha = 2H, \quad \beta = 1, \quad \sigma_*^2 = T^{2H}.$$

If $H \leq 1/2$, the matching increment lower bound applies; if $H \geq 1/2$, the matching variance-loss lower bound applies. Thus $\gamma = \gamma_H$ in Proposition 2.2.

When $H > 1/2$, Theorem 2.4 gives the exact Gaussian equivalent for every continuous trend. At $H = 1/2$, Example 3.1 gives the limiting ratio $\sqrt{2}$ in the zero-trend case. When $H < 1/2$, (34) gives the logarithmic expansion; an exact tail equivalent in this regime is not established here.

For growing domains take $m \equiv 0$ and suppose that $T_b \rightarrow \infty$ and $L_b := \log(b/T_b) \rightarrow \infty$ as $b \rightarrow \infty$. Since B_H is H -self-similar with $\sigma_1^2 = 1$, Proposition 2.10 yields

$$\log \mathbb{P} \left\{ \int_0^{T_b} e^{B_H(t)} dt > b \right\} \sim -\frac{L_b^2}{2T_b^{2H}} \quad \text{if } T_b^H = o(L_b).$$

If $T_b^{2H} = o(L_b)$ then for every $H \in (0, 1)$,

$$(35) \quad \log \mathbb{P} \left\{ \int_0^{T_b} e^{B_H(t)} dt > b \right\} = -\frac{L_b^2}{2T_b^{2H}} - \frac{L_b \log L_b}{\gamma_H T_b^{2H}} + O\left(\frac{L_b}{T_b^{2H}} + \log L_b\right).$$

If $H > 1/2$ and

$$T_b^{2H} = O(L_b),$$

then

$$(36) \quad \mathbb{P} \left\{ \int_0^{T_b} e^{B_H(t)} dt > b \right\} \sim \Psi(\rho_{\mathcal{A}_{T_b}, 0}^{B_H}(b)).$$

Example 3.4. Let $X(t), t \in [0, T]$ be a centred Gaussian process with continuous sample paths and stationary increments, and suppose that $X(0) = 0$ almost surely and write v for the variance function. Assume that $v \in C^1([0, T])$, that $v(t) < v(T)$ for every $t < T$, that

$$v'(T) > 0$$

and that, for some $\alpha \in (1, 2]$ and $\varepsilon > 0$

$$v(h) \leq Ch^\alpha, \quad 0 \leq h \leq \varepsilon.$$

Then the conclusions of Theorem 2.4 hold.

Indeed, stationary increments imply

$$\mathbb{E}\{(X(t) - X(s))^2\} = v(|t - s|) \leq C|t - s|^\alpha$$

for s, t sufficiently close to T . The assumptions on the variance function v therefore allow us to apply Corollary 2.5.

In particular, this example includes the scaled fractional Brownian motion $X(t) = cB_H(t)$ with $c \neq 0$ and $H \in (1/2, 1)$. It also includes

$$X(t) = c_1 B_{H_1}^{(1)}(t) + c_2 B_{H_2}^{(2)}(t),$$

where $B_{H_1}^{(1)}$ and $B_{H_2}^{(2)}$ are independent fractional Brownian motions, $H_1, H_2 \in (1/2, 1)$, and $(c_1, c_2) \neq (0, 0)$.

Example 3.5. Let $\mathcal{W}$ be an isonormal Gaussian measure on $L^2([0, T])$. Let $H \in (1/2, 1)$ and $\ell \in C^1([0, T])$ satisfy $\ell(T) \neq 0$. Put

$$k(u) := u^{H-1/2} \ell(u), \quad X_\ell(t) := \int_0^t k(t-s) \mathcal{W}(ds), \quad t \in [0, T].$$

Then X_ℓ has a continuous version and the conclusions of Theorem 2.4 hold. Indeed, with

$$q(t) := \int_0^t k(u)^2 du,$$

we have

$$v(t) = q(t), \quad v(t) < v(T) \quad (t < T), \quad v'(T) = k(T)^2 > 0.$$

To control the canonical metric, put

$$K_t(u) := k(t - u)\mathbf{1}_{[0,t]}(u).$$

If $h = t - s > 0$ and $a = H - 1/2$, then

$$\begin{aligned} \|K_t - K_s\|_2^2 &= \int_0^s \{k(x + h) - k(x)\}^2 dx + \int_0^h k(x)^2 dx \\ &\leq C \int_0^s \{(x + h)^a - x^a\}^2 dx + Ch^2 \int_0^s x^{2a} dx + Ch^{2H} \\ &\leq Ch^{2H}. \end{aligned}$$

Here the last estimate follows by substituting $x = hy$ in the first integral and using $2a - 2 < -1$. Consequently,

$$\mathbb{E}\{(X_\ell(t) - X_\ell(s))^2\} = \|K_t - K_s\|_2^2 \leq C|t - s|^{2H}.$$

Gaussian moment equivalence and the Kolmogorov criterion give a continuous version. Since $2H > 1$, [Corollary 2.5](#) applies.

Remark 3.6. Taking $\ell \equiv \sqrt{2H}$ gives the normalised Riemann–Liouville fractional Brownian motion. Thus this process is covered by the preceding example. Standard fractional Brownian motion is not a member of this convolution Volterra family.

Example 3.7. Let $d \geq 2$, $D = [0, T]^d$, and let B_H be the centred Gaussian field with covariance

$$R(s, t) = \prod_{i=1}^d \frac{s_i^{2H_i} + t_i^{2H_i} - |s_i - t_i|^{2H_i}}{2}, \quad H_i \in (1/2, 1), \quad i = 1, \dots, d.$$

Its variance is

$$v(t) = \prod_{i=1}^d t_i^{2H_i}, \quad \sigma_D^2 = T^{2\sum_i H_i}.$$

The unique variance maximum is at $\mathbf{T}$. Writing $x_i = t_i/T$, the inequalities

$$\max_{1 \leq i \leq d} (1 - x_i^{2H_i}) \leq 1 - \prod_{1 \leq i \leq d} x_i^{2H_i} \leq \sum_{1 \leq i \leq d} (1 - x_i^{2H_i})$$

and $1 - x^{2H_i} \asymp 1 - x$ on $[0, 1]$ give $\sigma_D^2 - v(t) \asymp \sum_{1 \leq i \leq d} (T - t_i)$. Telescoping one coordinate at a time and applying Cauchy–Schwarz give

$$\mathbb{E}\{(B_H(s) - B_H(t))^2\} \leq d \sum_{i=1}^d T^{2\sum_{j \neq i} H_j} |s_i - t_i|^{2H_i}.$$

This bound also gives a continuous version by Gaussian moment equivalence and the multiparameter Kolmogorov criterion. Thus [Remark 2.7](#) applies, for every continuous trend, with $\beta_i = 1$, $\alpha_i = 2H_i$ and $\gamma_\beta = d$. The requirement $H_i > 1/2$ in every coordinate is essential for this application: a coordinate with $H_i = 1/2$ fails the strict scale separation. In particular, the standard Brownian sheet is not covered.

4. PROOFS OF THE MAIN RESULTS

The proofs follow the order of Section 2. The variational notation and bounds used throughout are in [Appendix A.1](#); the Gaussian regression and short-interval estimate are in [Appendix A.2](#), and the absolute-continuity lemma is in [Appendix A.3](#).

Proof of Lemma 2.1. The upper bound is (68) with $A = \mathcal{A}_T$. Since $D_{\mathcal{A}_T, m}(f) \geq 0$ and $V_{\mathcal{A}_T}(f) \leq \sigma_*^2$, the variational formula gives

$$(37) \quad \rho_{\mathcal{A}_T, m}(b) \geq \frac{L_b}{\sigma_*}.$$

Choose $t_* \in \mathcal{A}_T$ with $v(t_*) = \sigma_*^2$ and fix $\varepsilon \in (0, \sigma_*^2)$. Continuity of R gives a relative interval $U_\varepsilon \subset \mathcal{A}_T$ containing t_* , of positive length, with $R(s, t) \geq \sigma_*^2 - \varepsilon$ on U_ε^2 . Set

$$C_\varepsilon := \int_{U_\varepsilon} e^{m(t)} dt, \quad q_\varepsilon(t) := \frac{e^{m(t)} \mathbf{1}_{U_\varepsilon}(t)}{C_\varepsilon}.$$

Then $q_\varepsilon \in \mathcal{D}(\mathcal{A}_T)$, $V_{\mathcal{A}_T}(q_\varepsilon) \geq \sigma_*^2 - \varepsilon > 0$, and $D_{\mathcal{A}_T, m}(q_\varepsilon) = \log\{M_m(\mathcal{A}_T)/C_\varepsilon\}$. Testing (60) with q_ε therefore yields

$$\rho_{\mathcal{A}_T, m}(b) \leq \frac{\log(b/C_\varepsilon)}{\sqrt{\sigma_*^2 - \varepsilon}}.$$

Letting first $b \rightarrow \infty$ and then $\varepsilon \downarrow 0$ proves $\rho_{\mathcal{A}_T, m}(b) \sim L_b/\sigma_*$. The upper bound and (2), together with Mills' formula, give

$$-\frac{\rho_{\mathcal{A}_T, m}(b)^2}{2} + O(\log \rho_{\mathcal{A}_T, m}(b)) \leq \log p_T(b) \leq -\frac{L_b^2}{2\sigma_*^2}.$$

The claimed equivalents follow, since $L_b \sim \log b$. □

Proof of Proposition 2.2. Put $\gamma := \min\{\alpha, \beta\}$ and write

$$I := I_{X, m}(\mathcal{A}_T), \quad d_X(s, t)^2 := \mathbb{E}\{(X(s) - X(t))^2\}, \quad q(t) := \sigma_*^2 - v(t).$$

For every $n \in \mathbb{N}$, Fubini's theorem and the Gaussian moment-generating formula yield

$$(38) \quad \mathbb{E}\{I^n\} = \int_{[0, T]^n} \exp\left\{\sum_{i=1}^n m(t_i) + \frac{1}{2} \sum_{i, j=1}^n R(t_i, t_j)\right\} dt_1 \cdots dt_n.$$

Since $2R(s, t) = v(s) + v(t) - d_X(s, t)^2$, we have the exact identity

$$(39) \quad \frac{1}{2} \sum_{i, j=1}^n R(t_i, t_j) = \frac{\sigma_*^2 n^2}{2} - \frac{n}{2} \sum_{i=1}^n q(t_i) - \frac{1}{2} \sum_{1 \leq i < j \leq n} d_X(t_i, t_j)^2.$$

We first prove a moment upper bound. Suppose that (10) holds. Put

$$\Gamma(s, t) := q(s) + q(t) + d_X(s, t)^2, \quad H_n(t_1, \dots, t_n) := \frac{1}{4} \sum_{i, j=1}^n \Gamma(t_i, t_j),$$

and

$$J_n := \int_{\mathcal{A}_T^n} e^{-H_n(t_1, \dots, t_n)} dt_1 \cdots dt_n.$$

By (39) we have

$$\mathbb{E}\{I^n\} \leq \exp\left\{\frac{\sigma_*^2 n^2}{2} + n\|m\|_\infty\right\} J_n.$$

Choose $\varepsilon_0 := \min\{\varepsilon/2, T/2\}$, set $U := [T - \varepsilon_0, T]$, and let

$$\eta := \min_{t \in [0, T - \varepsilon_0]} q(t) > 0.$$

For $x \in \mathcal{A}_T$, define

$$B_n(x) := \int_0^T \exp\left\{-\frac{n}{4}\Gamma(t, x)\right\} dt.$$

If $x \notin U$, then $B_n(x) \leq T e^{-n\eta/4}$. If $x \in U$, splitting the integral over U and its complement and using (10) implies

$$B_n(x) \leq \int_U e^{-c_d n|t-x|^\alpha/4} dt + T e^{-n\eta/4} \leq C n^{-1/\alpha}$$

and hence $\sup_{x \in \mathcal{A}_T} B_n(x) \leq C n^{-1/\alpha}$. For $D_j(t_1, \dots, t_n) := \sum_{i=1}^n \Gamma(t_i, t_j)$, convexity gives

$$e^{-H_n} = \exp\left\{\frac{1}{n} \sum_{j=1}^n -\frac{n}{4} D_j\right\} \leq \frac{1}{n} \sum_{j=1}^n e^{-n D_j/4}.$$

Integrating first in the variables different from t_j , using $\Gamma(x, x) = 2q(x)$ and symmetry, yields

$$J_n \leq \int_0^T e^{-nq(x)/2} B_n(x)^{n-1} dx \leq T(Cn^{-1/\alpha})^{n-1}.$$

Since $\gamma = \alpha$ in this case, this gives the desired upper bound.

Suppose next that (11) holds. By continuity of q and uniqueness of its zero at T , after decreasing c_v if necessary its lower bound extends to all $t \in \mathcal{A}_T$. The moment bound (67) therefore implies

$$\mathbb{E}\{I^n\} \leq \exp\left\{\frac{\sigma_*^2 n^2}{2}\right\} \left(\int_0^T e^{m(t)-nq(t)/2} dt\right)^n \leq \exp\left\{\frac{\sigma_*^2 n^2}{2} + Cn\right\} n^{-n/\beta}.$$

Here $\gamma = \beta$. In either case

$$(40) \quad \log \mathbb{E}\{I^n\} \leq \frac{\sigma_*^2 n^2}{2} - \frac{n}{\gamma} \log n + Cn.$$

We turn to the tail asymptotics. For all sufficiently large b , let

$$n_b := \left\lfloor \frac{L_b}{\sigma_*^2} \right\rfloor.$$

By Markov's inequality and (40)

$$\begin{aligned} \log \mathbb{P}\{I > b\} &\leq \log \mathbb{E}\{I^{n_b}\} - n_b \log b \\ &\leq \frac{\sigma_*^2 n_b^2}{2} - n_b L_b - \frac{n_b}{\gamma} \log n_b + O(n_b) \leq -\frac{L_b^2}{2\sigma_*^2} - \frac{L_b \log L_b}{\gamma \sigma_*^2} + CL_b. \end{aligned}$$

Here the constant term $\log M_m(\mathcal{A}_T)$ has been absorbed into $O(n_b)$.

For the lower bound set

$$\delta_b := L_b^{-1/\gamma}, \quad Q_b := [T - \delta_b, T],$$

and define

$$Y_b := \frac{1}{\delta_b} \int_{Q_b} X(t) dt, \quad \bar{m}_b := \frac{1}{\delta_b} \int_{Q_b} m(t) dt.$$

For all sufficiently large b , $Q_b \subset [T - \varepsilon, T]$. The random variable Y_b is centred Gaussian, and Jensen's inequality implies

$$(41) \quad I \geq \int_{Q_b} e^{X(t)+m(t)} dt \geq \delta_b e^{Y_b + \bar{m}_b}.$$

Writing $\tilde{V}_b := \text{Var}(Y_b)$, we have the exact identity

$$(42) \quad \sigma_*^2 - \tilde{V}_b = \frac{1}{\delta_b} \int_{Q_b} q(t) dt + \frac{1}{2\delta_b^2} \int_{Q_b} \int_{Q_b} d_X(s, t)^2 ds dt.$$

Consequently, we have

$$0 \leq \sigma_*^2 - \tilde{V}_b \leq C(\delta_b^\beta + \delta_b^\alpha) = O(L_b^{-1}),$$

and in particular $\tilde{V}_b > 0$ for all sufficiently large b . By continuity of the trend m

$$\bar{m}_b = m(T) + o(1)$$

and hence

$$z_b := \log \frac{b}{\delta_b} - \bar{m}_b = L_b + \frac{1}{\gamma} \log L_b + O(1).$$

Therefore by (41) we obtain

$$\mathbb{P}\{I > b\} \geq \mathbb{P}\{Y_b > z_b\} = \Psi\left(\frac{z_b}{\sqrt{\tilde{V}_b}}\right).$$

Mills' inequality now implies for some positive finite constant C

$$\log \mathbb{P}\{I > b\} \geq -\frac{z_b^2}{2\tilde{V}_b} - C \log z_b.$$

Using (42) and the expansion of z_b , we obtain

$$\frac{z_b^2}{2\tilde{V}_b} = \frac{L_b^2}{2\sigma_*^2} + \frac{L_b \log L_b}{\gamma\sigma_*^2} + O(L_b).$$

It follows that

$$\log \mathbb{P}\{I > b\} \geq -\frac{L_b^2}{2\sigma_*^2} - \frac{L_b \log L_b}{\gamma\sigma_*^2} - CL_b.$$

Combining the upper and lower bounds proves (12). $\square$

Proof of Corollary 2.3. Write $L = L_b$, $s = \sigma_*^2$, and $\rho_b = \rho_{\mathcal{A}_T, m}(b)$. Use the interval Q_b , its length δ_b , and the quantities $z_b, \tilde{V}_b$ from the preceding proof. The probability density $p_b := \delta_b^{-1} \mathbf{1}_{Q_b}$ has variance $V_{\mathcal{A}_T}(p_b) = \tilde{V}_b > 0$ for all sufficiently large b , and

$$\log b + \text{Ent}_{\mathcal{A}_T}(p_b) - \int_{\mathcal{A}_T} m(t)p_b(t) dt = z_b.$$

Testing (60) with p_b and using the expansion already established gives

$$\rho_b^2 \leq \frac{z_b^2}{\tilde{V}_b} = \frac{L^2}{s} + \frac{2}{\gamma s} L \log L + O(L).$$

For the reverse bound, put $I := I_{X, m}(\mathcal{A}_T)$. For $h \in \mathcal{H}_X(\mathcal{A}_T)$ and $n \in \mathbb{N}$, the reproducing property and Young's inequality give

$$\sum_{i=1}^n h(t_i) \leq \frac{1}{2} \|h\|_{\mathcal{H}_X(\mathcal{A}_T)}^2 + \frac{1}{2} \sum_{i,j=1}^n R(t_i, t_j).$$

Exponentiating and integrating against $\exp\{\sum_i m(t_i)\} dt$, and using (38), yields

$$\left(\int_{\mathcal{A}_T} e^{m(t)+h(t)} dt \right)^n \leq e^{\|h\|_{\mathcal{H}_X(\mathcal{A}_T)}^2/2} \mathbb{E}\{I^n\}.$$

Taking the infimum over feasible h therefore gives

$$\frac{\rho_b^2}{2} \geq n \log b - \log \mathbb{E}\{I^n\}.$$

Choose $n = \lfloor L/s \rfloor$. By (40), for all sufficiently large b

$$\frac{\rho_b^2}{2} \geq \frac{L^2}{2s} + \frac{L \log L}{\gamma s} - CL,$$

which together with the upper bound proves (13). $\square$

Proof of Theorem 2.4. Set next

$$L_b := \log \frac{b}{M_m(\mathcal{A}_T)}, \quad \rho_b := \rho_{\mathcal{A}_T, m}(b).$$

For all sufficiently large b , define

$$\varepsilon_b := \frac{(\log L_b)^2}{L_b}, \quad r_b := \varepsilon_b^{1/\beta}, \quad \delta_b := L_b^{-2},$$

and

$$Q_b := [T - r_b, T], \quad A_b := [0, T - r_b].$$

We prove that as $b \rightarrow \infty$

$$(43) \quad \mathbb{P}\{I_{X, m}(Q_b) > (1 - \delta_b)b\} \leq (1 + o(1))\Psi(\rho_b),$$

$$(44) \quad \mathbb{P}\{I_{X, m}(A_b) > \delta_b b\} = o(\Psi(\rho_b)).$$

Write

$$L := L_b, \quad s := \sigma_*^2.$$

By continuity of v , uniqueness of its maximum at T , and Assumption (C1), after decreasing its constant if necessary we may assume that

$$(45) \quad s - v(t) \geq c(T - t)^\beta, \quad t \in \mathcal{A}_T.$$

Setting

$$\chi_b := r_b^\alpha = \varepsilon_b^{\alpha/\beta}$$

and recalling that $\alpha > \beta$ we have as $b \rightarrow \infty$

$$(46) \quad L\chi_b = L^{1-\alpha/\beta}(\log L)^{2\alpha/\beta} \rightarrow 0, \quad \frac{L\varepsilon_b}{\log L} = \log L \rightarrow \infty.$$

We first consider the endpoint interval. Define

$$\hat{\rho}_b := \rho_{Q_b, m}((1 - \delta_b)b), \quad \hat{L}_b := \log \frac{(1 - \delta_b)b}{M_m(Q_b)}.$$

These quantities are well defined for all sufficiently large b . Indeed, $r_b \rightarrow 0$ as $b \rightarrow \infty$ and continuity of m gives

$$M_m(Q_b) \asymp r_b, \quad \hat{L}_b = L + O(\log L).$$

Continuity of R at (T, T) also gives, for all sufficiently large b ,

$$R(u, t) \geq \frac{s}{2}, \quad u, t \in Q_b.$$

Moreover,

$$R(t, t) \leq s, \quad \mathbb{E}\{(X(u) - X(t))^2\} \leq C_d |u - t|^\alpha, \quad u, t \in Q_b.$$

Since $\hat{L}_b = L + O(\log L)$, (46) implies

$$(1 + \hat{L}_b)\chi_b \rightarrow 0, \quad b \rightarrow \infty.$$

Hence Lemma A.6 with $\tau_b = 1$ gives

$$(47) \quad \mathbb{P}\{I_{X, m}(Q_b) > (1 - \delta_b)b\} \leq (1 + o(1))\Psi(\hat{\rho}_b).$$

We next compare the local and global variational levels. Let $\hat{f}_b$ be a minimiser in the variational formula defining $\hat{\rho}_b$, and extend it by zero from Q_b to $\mathcal{A}_T$. Since $R(u, t) \geq s/2$ on Q_b^2 ,

$$V_{Q_b}(\hat{f}_b) \geq \frac{s}{2}.$$

Put

$$\eta_b := -\log(1 - \delta_b) = O(L^{-2}).$$

Using the extended density as a competitor in the global variational problem gives

$$\rho_b \leq \frac{\log b + \text{Ent}_{Q_b}(\hat{f}_b) - \int_{Q_b} m(t)\hat{f}_b(t) dt}{\sqrt{V_{Q_b}(\hat{f}_b)}} = \hat{\rho}_b + \frac{\eta_b}{\sqrt{V_{Q_b}(\hat{f}_b)}} \leq \hat{\rho}_b + C\eta_b$$

implying

$$(48) \quad \hat{\rho}_b \geq \rho_b - C\eta_b.$$

By Lemma 2.1, $\rho_b = O(L)$. The standard normal hazard bound therefore yields

$$\frac{\Psi(\hat{\rho}_b)}{\Psi(\rho_b)} \leq \frac{\Psi(\rho_b - C\eta_b)}{\Psi(\rho_b)} \leq \exp\{C\eta_b(1 + \rho_b)\} = 1 + o(1),$$

which together with (47) proves (43).

It remains to control the complement. Set next

$$s_{A, b}^2 := \max_{t \in A_b} v(t), \quad L_{A, b} := \log \frac{\delta_b b}{M_m(A_b)}.$$

In view of (45) we have

$$s_{A, b}^2 \leq s - cr_b^\beta = s - c\varepsilon_b.$$

Furthermore, since $M_m(A_b) \leq M_m(\mathcal{A}_T)$

$$L_{A,b} \geq L - 2 \log L.$$

In particular, $L_{A,b} \geq s_{A,b}^2$ for all sufficiently large b . The moment bound in [Lemma A.2](#), applied on A_b , therefore yields

$$\begin{aligned} \log \mathbb{P}\{I_{X,m}(A_b) > \delta_b b\} &\leq -\frac{L_{A,b}^2}{2s_{A,b}^2} \\ &\leq -\frac{(L - 2 \log L)^2}{2(s - c\varepsilon_b)} \\ (49) \quad &\leq -\frac{L^2}{2s} - c_1 L^2 \varepsilon_b + CL \log L \end{aligned}$$

for some $c_1 > 0$.

On the other hand, [Corollary 2.3](#), applied with $\gamma = \beta$, implies

$$\rho_b^2 \leq \frac{L^2}{s} + CL \log L.$$

Mills' lower bound and $\rho_b = O(L)$ consequently give

$$-\log \Psi(\rho_b) \leq \frac{L^2}{2s} + CL \log L.$$

Combining this with (49) and using $L^2 \varepsilon_b = L(\log L)^2$, we obtain

$$\log \frac{\mathbb{P}\{I_{X,m}(A_b) > \delta_b b\}}{\Psi(\rho_b)} \leq -c_1 L(\log L)^2 + CL \log L \longrightarrow -\infty, \quad b \rightarrow \infty$$

establishing (44). Finally, the common endpoint of A_b and Q_b is Lebesgue-null, so

$$I_{X,m}(\mathcal{A}_T) = I_{X,m}(A_b) + I_{X,m}(Q_b)$$

and therefore

$$\{I_{X,m}(\mathcal{A}_T) > b\} \subset \{I_{X,m}(Q_b) > (1 - \delta_b)b\} \cup \{I_{X,m}(A_b) > \delta_b b\}.$$

The union bound and (43)–(44) yield

$$(50) \quad \mathbb{P}\{I_{X,m}(\mathcal{A}_T) > b\} \leq (1 + o(1))\Psi(\rho_b).$$

The reverse inequality is (2), so (15) follows.

The expansion (16) follows from [Corollary 2.3](#) with $\gamma = \beta$, since Assumption (C1) supplies the matching variance-loss lower bound. □

Proof of Corollary 2.5. The expansion $\sigma_*^2 - v(T - h) = v'(T)h + o(h)$ as $h \downarrow 0$ gives Assumption (C1) with $\beta = 1$, so [Theorem 2.4](#) applies. □

Proof of Theorem 2.8. We use the concavity of the Gaussian quantile transform, as in the density analysis of [7], and verify the argument under the present assumptions. Applying [Lemma A.7](#) to the Gaussian process $X + m$ shows that I is absolutely continuous under the standing assumptions. Since $I > 0$ almost surely, the change of variables $Y = \log I$ also gives absolute continuity of Y .

Set

$$\rho_b := \rho_{\mathcal{A}_T, m}(b), \quad \overline{F}_Y(B) := 1 - F_Y(B) = \mathbb{P}\{Y > B\}.$$

In view of [Lemma 2.1](#) we have

$$\log \overline{F}_Y(B) \sim -\frac{B^2}{2\sigma_*^2}, \quad \rho_{e^B} \sim \frac{B}{\sigma_*}, \quad B \rightarrow \infty.$$

Choose $B_* > 0$ sufficiently large that $0 < F_Y(B) < 1$ for $B > B_*$, and put

$$\Theta(B) := \Phi^{-1}(F_Y(B)), \quad B > B_*.$$

Thus Θ is finite and $\Psi(\Theta(B)) = \overline{F}_Y(B)$.

Define the continuous convex functional

$$\mathcal{L}(x) := \log \int_{\mathcal{A}_T} e^{x(t)+m(t)} dt, \quad x \in C(\mathcal{A}_T),$$

and its closed sublevel sets $A_B := \{x : \mathcal{L}(x) \leq B\}$. Indeed, $|\mathcal{L}(x) - \mathcal{L}(y)| \leq \|x - y\|_\infty$, and Hölder's inequality gives

$$(1 - \lambda)A_{B_1} + \lambda A_{B_2} \subset A_{(1-\lambda)B_1 + \lambda B_2}, \quad 0 < \lambda < 1.$$

Let μ_X be the law of X on the separable Banach space $C(\mathcal{A}_T)$. It is a Gaussian Radon measure. Write

$$\mu_{X,*}(C) := \sup\{\mu_X(K) : K \subset C \text{ compact}\}.$$

The inner-measure form of Ehrhard's inequality [19, Theorem 3.1] therefore yields, for $B_1, B_2 > B_*$,

$$\begin{aligned} \Theta((1 - \lambda)B_1 + \lambda B_2) &\geq \Phi^{-1}(\mu_{X,*}((1 - \lambda)A_{B_1} + \lambda A_{B_2})) \\ &\geq (1 - \lambda)\Phi^{-1}(\mu_X(A_{B_1})) + \lambda\Phi^{-1}(\mu_X(A_{B_2})) \\ &= (1 - \lambda)\Theta(B_1) + \lambda\Theta(B_2). \end{aligned}$$

Hence Θ is concave on (B_*, ∞) and has a finite right derivative there.

The logarithmic tail asymptotics and $\Psi(\Theta(B)) = \bar{F}_Y(B)$ imply

$$\Theta(B) \sim \rho_{e^B} \sim \frac{B}{\sigma_*}, \quad B \rightarrow \infty.$$

For $B > 2B_*$, concavity gives

$$\frac{\Theta(2B) - \Theta(B)}{B} \leq \Theta'_+(B) \leq \frac{\Theta(B) - \Theta(B/2)}{B/2}.$$

Both secants tend to $1/\sigma_*$ as $B \rightarrow \infty$, so

$$\Theta'_+(B) \rightarrow \frac{1}{\sigma_*}, \quad B \rightarrow \infty.$$

Since $F_Y = \Phi \circ \Theta$, its right derivative exists on (B_*, ∞) , and Mills' formula gives

$$\begin{aligned} (51) \quad F'_{Y,+}(B) &= \varphi(\Theta(B))\Theta'_+(B) \\ &\sim \frac{\Theta(B)}{\sigma_*}\Psi(\Theta(B)) \\ &\sim \frac{B}{\sigma_*^2}\bar{F}_Y(B) \sim \frac{\rho_{e^B}}{\sigma_*}\bar{F}_Y(B). \end{aligned}$$

This derivative is nonnegative and measurable: Θ is continuous and Θ'_+ is monotone. Let $\tilde{g}_Y$ be any density of Y and define

$$g_Y(B) := \begin{cases} F'_{Y,+}(B), & B > B_*, \\ \tilde{g}_Y(B), & B \leq B_*. \end{cases}$$

Absolute continuity of F_Y implies that g_Y agrees almost everywhere with $\tilde{g}_Y$, so g_Y is a version of the density of Y . Therefore

$$f_I(b) := \frac{1}{b}g_Y(\log b), \quad b > 0,$$

is a version of the density of I and satisfies (21) for $b > e^{B_*}$. Equation (51) proves (22). $\square$

Proof of Corollary 2.9. Put $\rho_b := \rho_{\mathcal{A}_T, m}(b)$ and take the density version from Theorem 2.8. By Theorem 2.4, $\mathbb{P}\{I > b\} \sim \Psi(\rho_b)$. Hence (22) and Mills' formula yield

$$f_I(b) \sim \frac{\rho_b \Psi(\rho_b)}{\sigma_* b} \sim \frac{\varphi(\rho_b)}{\sigma_* b},$$

which is (23). $\square$

Proof of Proposition 2.10. By self-similarity

$$(52) \quad X(0) = 0 \quad \text{a.s.}, \quad v(t) = \sigma_1^2 t^{2\kappa}, \quad t \geq 0,$$

and for all $b > T > 0$

$$(53) \quad \int_0^T e^{X(t)} dt \stackrel{d}{=} T \int_0^1 e^{T^\kappa X(s)} ds,$$

$$(54) \quad \rho_{\mathcal{A}_T,0}^X(b) = \rho_{\mathcal{A}_1,0}^{T^\kappa X}(b/T) = T^{-\kappa} \rho_{\mathcal{A}_1,0}^X(b/T).$$

The level identities follow from the time change $t = Ts$ and the RKHS scaling $\mathcal{H}_{aX} = \mathcal{H}_X$, $\|h\|_{\mathcal{H}_{aX}} = a^{-1} \|h\|_{\mathcal{H}_X}$ for $a > 0$.

Write

$$a := T_b^\kappa, \quad L := L_b, \quad J_a := \int_0^1 e^{aX(t)} dt, \quad \rho := \rho_{\mathcal{A}_{T_b},0}(b).$$

The preceding identities give

$$p_{T_b}(b) = \mathbb{P}\{J_a > e^L\}, \quad \rho = a^{-1} \rho_{\mathcal{A}_1,0}^X(e^L).$$

Self-similarity also gives

$$q(t) := \sigma_1^2 - v(t) = \sigma_1^2(1 - t^{2\kappa}) \asymp 1 - t, \quad t \uparrow 1,$$

and $q(t) \geq c(1 - t)$ on $[0, 1]$ for some $c > 0$. For part (i), Lemma 2.1 applied on $[0, 1]$ implies

$$\rho \sim \frac{L}{a\sigma_1} \longrightarrow \infty, \quad b \rightarrow \infty.$$

The variational lower bound and Mills' formula give the logarithmic lower estimate. Put

$$\mu_1 := \mathbb{E}\left\{\sup_{0 \leq t \leq 1} X(t)\right\} < \infty.$$

Borell–TIS gives for all sufficiently large b

$$p_{T_b}(b) \leq \exp \left\{ -\frac{1}{2\sigma_1^2} \left(\frac{L}{a} - \mu_1 \right)^2 \right\}.$$

The estimates match since $L/a \rightarrow \infty$ as $b \rightarrow \infty$. Moreover, $\log T_b = o(T_b^\kappa) = o(L)$, so $L \sim \log b$. This proves (26).

For part (ii), Lemma A.3 applies with $\gamma = \min\{\alpha, 1\}$. Since $a^2 = o(L)$, the exponent $\lambda := L/(\sigma_1^2 a^2)$ tends to infinity as $b \rightarrow \infty$. Markov's inequality and (70) imply

$$(55) \quad \log p_{T_b}(b) \leq -\frac{L^2}{2\sigma_1^2 a^2} - \frac{L \log L}{\gamma \sigma_1^2 a^2} + O\left(\frac{L}{a^2} + \log L\right).$$

For the lower bound put

$$\delta := L^{-1/\gamma}, \quad Q := [1 - \delta, 1], \quad Y_Q := \frac{1}{\delta} \int_Q X(t) dt, \quad V_Q := \text{Var}(Y_Q).$$

For all sufficiently large b , $Q \subset [1 - \varepsilon, 1]$. The variance-deficit identity gives

$$\sigma_1^2 - V_Q = \frac{1}{\delta} \int_Q q(t) dt + \frac{1}{2\delta^2} \int_Q \int_Q \mathbb{E}\{(X(s) - X(t))^2\} ds dt = O(\delta + \delta^\alpha) = O(L^{-1}).$$

In particular $V_Q \geq \sigma_1^2/2$ eventually and $V_Q^{-1} = \sigma_1^{-2} + O(L^{-1})$. Jensen's inequality yields $J_a \geq \delta e^{aY_Q}$ and hence

$$p_{T_b}(b) \geq \Psi \left(\frac{L + \gamma^{-1} \log L}{a\sqrt{V_Q}} \right).$$

The Gaussian level tends to infinity as $b \rightarrow \infty$. Expanding its square and using Mills' formula gives

$$\log p_{T_b}(b) \geq -\frac{L^2}{2\sigma_1^2 a^2} - \frac{L \log L}{\gamma \sigma_1^2 a^2} - O\left(\frac{L}{a^2} + \log L\right).$$

Together with (55) this proves (27).

For part (iii), the variance-loss exponent on $[0, 1]$ is $\beta = 1$ and $\alpha > 1$. Therefore Corollary 2.3 and (54) give (30). By (28), $a^2 = O(L)$, and consequently $\rho \sim L/(a\sigma_1) \rightarrow \infty$ as $b \rightarrow \infty$. Set

$$r := \frac{(\log L)^2}{L}, \quad Q := [1 - r, 1], \quad \delta := L^{-2}, \quad \hat{c} := (1 - \delta)e^L, \quad \hat{\rho} := \rho_{Q,0}^{aX}(\hat{c}).$$

For large b , continuity of R gives $R(s, t) \geq \sigma_1^2/2$ on Q^2 . The covariance and increment bounds for aX satisfy Lemma A.6 with $\tau_b = a$, while

$$\hat{\ell} := \log(\hat{c}/|Q|) = L + O(\log L).$$

Since $a^2 = O(L)$ and $\alpha > 1$

$$(a + \hat{\ell})r^\alpha = O(L^{1-\alpha}(\log L)^{2\alpha}) \rightarrow 0, \quad b \rightarrow \infty.$$

The uniform short-interval estimate gives

$$(56) \quad \mathbb{P} \left\{ \int_Q e^{aX(t)} dt > (1 - \delta)e^L \right\} \leq (1 + o(1))\Psi(\hat{\rho}).$$

Extending a local minimising density by zero to $[0, 1]$ and using $V_Q^{aX}(f) \geq a^2\sigma_1^2/2$ gives

$$\hat{\rho} \geq \rho - \frac{C\eta}{a}, \quad \eta := -\log(1 - \delta) = O(L^{-2}).$$

Since $\rho = O(L/a)$, the normal hazard bound gives

$$\frac{\Psi(\hat{\rho})}{\Psi(\rho)} \leq \exp \left\{ \frac{C\eta}{a}(1 + \rho) \right\} = 1 + o(1).$$

For the complement $A := [0, 1 - r]$ write $I_{\text{out}} := \int_A e^{aX(t)} dt$ and put

$$u := \frac{L - 2\log L}{a} \rightarrow \infty, \quad b \rightarrow \infty.$$

Since $|A| \leq 1$

$$\{I_{\text{out}} > \delta e^L\} \subseteq \left\{ \sup_{t \in A} X(t) > u \right\}.$$

Fix $\varepsilon_0 \in (0, \min\{\varepsilon, 1\})$ and split A into

$$A_{\text{near}} := [1 - \varepsilon_0, 1 - r], \quad A_{\text{far}} := [0, 1 - \varepsilon_0].$$

For all sufficiently large b , cover A_{near} by at most $Cu^{2/\alpha}$ intervals $J_j \subset A_{\text{near}}$ of length at most $u^{-2/\alpha}$, and choose $t_j \in J_j$. The increment bound (24) and Dudley's bound give, uniformly in j ,

$$\mathbb{E}\{\sup_{t \in J_j} X(t)\} = \mathbb{E}\{\sup_{t \in J_j} \{X(t) - X(t_j)\}\} \leq C/u.$$

Moreover, $q(t) \geq c(1 - t)$ implies $\sup_{t \in A_{\text{near}}} v(t) \leq \sigma_1^2 - cr$. Borell–TIS and the union bound therefore yield

$$p_{\text{near}} := \mathbb{P} \left\{ \sup_{t \in A_{\text{near}}} X(t) > u \right\} \leq Cu^{2/\alpha} \exp \left\{ -\frac{u^2}{2(\sigma_1^2 - cr)} \right\},$$

where the constant absorbs the bounded correction from the means $\mathbb{E}\{\sup_{J_j} X\} \leq C/u$. The level expansion (30) and Mills' formula give

$$-\log \Psi(\rho) \leq \frac{L^2}{2\sigma_1^2 a^2} + C \frac{L \log L}{a^2} + C \log L.$$

Consequently,

$$\log \frac{p_{\text{near}}}{\Psi(\rho)} \leq -c' \frac{L(\log L)^2}{a^2} + C \frac{L \log L}{a^2} + C \log L \rightarrow -\infty, \quad b \rightarrow \infty.$$

Here L/a^2 is bounded away from zero, so the negative term dominates both positive terms.

On A_{far} we have

$$\sigma_0^2 := \sup_{t \in A_{\text{far}}} v(t) = \sigma_1^2(1 - \varepsilon_0)^{2\kappa} < \sigma_1^2, \quad \mu_0 := \mathbb{E} \left\{ \sup_{t \in A_{\text{far}}} X(t) \right\} < \infty.$$

A further application of Borell–TIS gives

$$p_{\text{far}} := \mathbb{P} \left\{ \sup_{t \in A_{\text{far}}} X(t) > u \right\} \leq \exp \left\{ -\frac{(u - \mu_0)^2}{2\sigma_0^2} \right\}.$$

Thus, for some $c'' > 0$

$$\log \frac{p_{\text{far}}}{\Psi(\rho)} \leq -c'' \frac{L^2}{a^2} + C \frac{L \log L}{a^2} + C \frac{L}{a} + C \log L \longrightarrow -\infty, \quad b \rightarrow \infty,$$

since $a^2 = O(L)$. It follows that

$$\mathbb{P}\{I_{\text{out}} > \delta e^L\} \leq p_{\text{near}} + p_{\text{far}} = o(\Psi(\rho)).$$

Combining this with (56), the local level comparison and the union bound proves the upper estimate in (29). The variational lower bound gives the reverse estimate. $\square$

APPENDIX A. AUXILIARY AND TECHNICAL RESULTS

A.1. Variational formulation and bounds. Let hereafter $X(t), t \in A$ be a centered Gaussian process with continuous sample paths where

$$A = [a^-, a^+] \subset \mathcal{A}_T, \quad a^- < a^+,$$

and assume the local nondegeneracy condition

$$(57) \quad v_A^* := \max_{t \in A} v(t) > 0.$$

For measurable $f : A \rightarrow [0, \infty)$ write

$$\text{Ent}_A(f) := \int_A f(t) \log f(t) dt, \quad 0 \log 0 := 0,$$

and define further

$$\mathcal{D}(A) := \left\{ f : A \rightarrow [0, \infty) \text{ measurable} : \int_A f(t) dt = 1, \quad \text{Ent}_A(f) < \infty \right\}.$$

Since $|A| < \infty$ and $x \log x \geq -e^{-1}$, the entropy is well defined in $(-\infty, \infty]$. Put

$$q_m(t) := \frac{e^{m(t)}}{M_m(A)}, \quad D_{A,m}(f) := \int_A f(t) \log \frac{f(t)}{q_m(t)} dt, \quad V_A(f) := \int_A \int_A R(s, t) f(s) f(t) ds dt.$$

Since q_m is bounded above and below by positive constants, $D_{A,m}(f) < \infty$ if and only if $\text{Ent}_A(f) < \infty$, and

$$(58) \quad D_{A,m}(f) = \text{Ent}_A(f) - \int_A m(t) f(t) dt + \log M_m(A) \geq 0.$$

Let $\mathcal{H}_X(A)$ be the RKHS of X on A . Every $h \in \mathcal{H}_X(A)$ is continuous and

$$|h(t)| \leq \|h\|_{\mathcal{H}_X(A)} \sqrt{v(t)}.$$

For $f \in \mathcal{D}(A)$, continuity of $s \mapsto R(\cdot, s)$ in the RKHS norm and boundedness of v show that the Bochner integral

$$K_A f := \int_A R(\cdot, s) f(s) ds$$

belongs to $\mathcal{H}_X(A)$ and satisfies

$$(59) \quad \langle h, K_A f \rangle_{\mathcal{H}_X(A)} = \int_A h(t) f(t) dt, \quad \|K_A f\|_{\mathcal{H}_X(A)}^2 = V_A(f).$$

For $b > M_m(A)$, put

$$L_{A,m}(b) := \log \frac{b}{M_m(A)} > 0.$$

If $m \equiv c$, then $\rho_{A,m}(b) = \rho_{A,0}(be^{-c})$.

The Gaussian-energy formulation and Gibbs stationarity are precedents from [13, Sections 2 and 3]; see also [14, Section 2]. The next lemma proves the full entropy identity and optimiser correspondence in the present setting, without a small-noise restriction.

Lemma A.1. *For every $b > M_m(A)$, the infimum in (3) is a positive finite minimum. Moreover,*

$$(60) \quad \rho_{A,m}(b) = \min_{\substack{f \in \mathcal{D}(A) \\ V_A(f) > 0}} \frac{\log b + \text{Ent}_A(f) - \int_A m(t)f(t) \, dt}{\sqrt{V_A(f)}}.$$

Every RKHS minimiser h satisfies

$$(61) \quad \int_A e^{m(t)+h(t)} \, dt = b.$$

For such an h , set

$$(62) \quad f(t) := \frac{e^{m(t)+h(t)}}{b}, \quad V := V_A(f), \quad g := \frac{K_A f}{\sqrt{V}}.$$

Then f is a strictly positive continuous probability density, $V > 0$, f is a minimiser in (60), and

$$(63) \quad \|g\|_{\mathcal{H}_X(A)} = 1, \quad h = \rho_{A,m}(b)g.$$

Conversely, if f is a minimiser in (60), then

$$h := \rho_{A,m}(b) \frac{K_A f}{\sqrt{V_A(f)}}$$

is an RKHS minimiser and $f = e^{m+h}/b$ almost everywhere. Thus the two minimiser sets are in bijection modulo equality almost everywhere.

Proof. Choose $t_0 \in A$ with $v(t_0) = v_A^* > 0$. By continuity of R , there are a relative neighbourhood U of t_0 in A and a constant $c_0 > 0$ such that

$$R(s, t) \geq c_0, \quad s, t \in U.$$

Since A is a nondegenerate interval, $|U| > 0$. Hence, for $f_0 := |U|^{-1} \mathbf{1}_U$,

$$f_0 \in \mathcal{D}(A), \quad V_A(f_0) \geq c_0 > 0.$$

For $h \in C(A)$, the Gibbs variational formula gives

$$(64) \quad \log \int_A q_m(t) e^{h(t)} \, dt = \sup_{f \in \mathcal{D}(A)} \left\{ \int_A h(t) f(t) \, dt - D_{A,m}(f) \right\}.$$

The supremum is attained uniquely at

$$f_h(t) = \frac{q_m(t) e^{h(t)}}{\int_A q_m(s) e^{h(s)} \, ds}.$$

Since m and h are bounded, f_h is a strictly positive continuous probability density and belongs to $\mathcal{D}(A)$. Because the supremum in (64) is attained, including when it is equal to $L_{A,m}(b)$, an element $h \in \mathcal{H}_X(A)$ is feasible in (3) if and only if it belongs to at least one of the half-spaces

$$H_f := \{h \in \mathcal{H}_X(A) : \langle h, K_A f \rangle_{\mathcal{H}_X(A)} \geq L_{A,m}(b) + D_{A,m}(f)\}, \quad f \in \mathcal{D}(A).$$

If $V_A(f) = 0$, then $K_A f = 0$ and $H_f = \emptyset$, since $L_{A,m}(b) + D_{A,m}(f) > 0$. If $V_A(f) > 0$, then

$$\text{dist}(0, H_f) = \frac{L_{A,m}(b) + D_{A,m}(f)}{\sqrt{V_A(f)}}.$$

Taking the infimum over these half-spaces and using

$$L_{A,m}(b) + D_{A,m}(f) = \log b + \text{Ent}_A(f) - \int_A m(t)f(t) dt$$

proves the variational identity (60) at the level of infima. The density f_0 constructed above shows that this common infimum is finite.

By (59) and the triangle inequality

$$\sqrt{V_A(f)} = \|K_A f\|_{\mathcal{H}_X(A)} \leq \int_A \sqrt{v(t)}f(t) dt \leq \sqrt{v_A^*}.$$

Since $D_{A,m}(f) \geq 0$, it follows that

$$(65) \quad \rho_{A,m}(b) \geq \frac{L_{A,m}(b)}{\sqrt{v_A^*}} > 0.$$

Let (h_n) be a minimising sequence in (3). It is norm-bounded, so weak sequential compactness of closed balls in a Hilbert space yields, after passage to a subsequence,

$$h_n \rightharpoonup h \quad \text{in } \mathcal{H}_X(A), \quad n \rightarrow \infty.$$

Point evaluations are continuous on $\mathcal{H}_X(A)$, and hence $h_n(t) \rightarrow h(t)$ as $n \rightarrow \infty$ for every $t \in A$. Furthermore,

$$\sup_{n,t} |h_n(t)| \leq \sup_n \|h_n\|_{\mathcal{H}_X(A)} \sqrt{v_A^*} < \infty.$$

Dominated convergence therefore preserves the constraint, while weak lower semicontinuity of the norm gives

$$\|h\|_{\mathcal{H}_X(A)} \leq \liminf_{n \rightarrow \infty} \|h_n\|_{\mathcal{H}_X(A)}.$$

Thus h is an RKHS minimiser.

Write $\rho := \rho_{A,m}(b)$. For $f \in \mathcal{D}(A)$ with $V_A(f) > 0$, put

$$N_f := L_{A,m}(b) + D_{A,m}(f), \quad p_f := \frac{N_f}{V_A(f)} K_A f.$$

The point p_f is the unique nearest point of H_f to zero.

Fix an RKHS minimiser h and take its Gibbs density

$$f(t) := \frac{e^{m(t)+h(t)}}{\int_A e^{m(s)+h(s)} ds}.$$

It is strictly positive and continuous and belongs to $\mathcal{D}(A)$. Gibbs' formula gives $h \in H_f$, so $V_A(f) > 0$. Since H_f is contained in the feasible set,

$$\rho \leq \|p_f\|_{\mathcal{H}_X(A)} \leq \|h\|_{\mathcal{H}_X(A)} = \rho.$$

Uniqueness of the nearest point in H_f implies $h = p_f$, and hence

$$(66) \quad \langle h, K_A f \rangle_{\mathcal{H}_X(A)} = N_f = L_{A,m}(b) + D_{A,m}(f).$$

Gibbs equality now gives

$$\log \frac{\int_A e^{m(t)+h(t)} dt}{M_m(A)} = \langle h, K_A f \rangle_{\mathcal{H}_X(A)} - D_{A,m}(f) = L_{A,m}(b).$$

Thus the constraint is active and $f = e^{m+h}/b$, proving (61). Moreover, $\rho = \|p_f\|_{\mathcal{H}_X(A)} = N_f/\sqrt{V_A(f)}$, so f attains the minimum in (60). With g as in (62), we have $h = p_f = \rho g$ and $\|g\|_{\mathcal{H}_X(A)} = 1$, proving (63).

Conversely, if f minimises the quotient in (60), then $p_f \in H_f$ is feasible and has norm ρ . Thus p_f is an RKHS minimiser and its constraint is active by the preceding argument. Consequently,

$$\langle p_f, K_A f \rangle_{\mathcal{H}_X(A)} - D_{A,m}(f) = L_{A,m}(b) = \log \int_A q_m(t) e^{p_f(t)} dt.$$

Equality in Gibbs' formula implies $f = e^{m+p_f}/b$ almost everywhere. This proves the asserted bijection between the two minimiser sets. $\square$

Lemma A.2. *For every $b > M_m(A)$, the lower bound (2) holds. Moreover, for all $r \geq 1$*

$$(67) \quad (\mathbb{E}\{I_{X,m}(A)^r\})^{1/r} \leq M_{m+(r/2)v}(A)$$

and for all $b \geq M_m(A)e^{v_A^}$, we have*

$$(68) \quad \mathbb{P}\{I_{X,m}(A) > b\} \leq \exp \left\{ -\frac{L_{A,m}(b)^2}{2v_A^*} \right\}.$$

Proof. By Lemma A.1, choose a strictly positive continuous minimising density f with $V_A(f) > 0$. Jensen's inequality gives

$$\log I_{X,m}(A) \geq \int_A (X(t) + m(t) - \log f(t)) f(t) dt.$$

The Gaussian random variable $\int_A X(t) f(t) dt$ is centred with variance $V_A(f)$ and hence

$$(69) \quad \mathbb{P}\{I_{X,m}(A) > b\} \geq \Psi \left(\frac{\log b + \text{Ent}_A(f) - \int_A m(t) f(t) dt}{\sqrt{V_A(f)}} \right) = \Psi(\rho_{A,m}(b)),$$

which is the Jensen–Gaussian bound of [3, Proposition 2.3] optimised using (60).

For $r \geq 1$, Minkowski's integral inequality and the Gaussian moment-generating formula yield

$$(\mathbb{E}\{I_{X,m}(A)^r\})^{1/r} \leq \int_A e^{m(t)} \left(\mathbb{E}\{e^{rX(t)}\} \right)^{1/r} dt = M_{m+(r/2)v}(A).$$

This proves (67); see also [3, Proposition 2.13(ii)]. The Jensen argument applies on compact rectangles, and the moment bound applies on measurable subsets of a compact parameter domain. Finally, writing $L = L_{A,m}(b)$, Markov's inequality yields

$$\mathbb{P}\{I_{X,m}(A) > b\} \leq \exp \left\{ -rL + \frac{r^2 v_A^*}{2} \right\}.$$

Choosing $r = L/v_A^* \geq 1$ proves (68). $\square$

We shall also need a moment estimate uniform in the amplitude.

Lemma A.3. *Let $X(t), t \in [0, 1]$ be a centred GP with almost surely continuous sample paths. Set*

$$\sigma_1^2 := v(1) > 0, \quad q(t) := \sigma_1^2 - v(t)$$

and suppose that $q(t) \geq c(1-t)$ for all $t \in [0, 1]$ and some $c > 0$. Let $\alpha \in (0, 2]$ and $\gamma := \min\{\alpha, 1\}$. If $\alpha < 1$ suppose further that for some $\varepsilon \in (0, 1]$ and $c_d > 0$ condition (25) holds. For $a \geq 1$ set

$$J_a := \int_0^1 e^{aX(t)} dt.$$

Then uniformly for $a \geq 1$ and $\lambda \geq 2$

$$(70) \quad \log \mathbb{E}\{J_a^\lambda\} \leq \frac{\sigma_1^2 a^2 \lambda^2}{2} - \frac{\lambda}{\gamma} \log(a^2 \lambda) + O(\lambda + \log(a^2 \lambda)).$$

Proof. For $n \in \mathbb{N}$ the Gaussian moment-generating formula gives

$$\mathbb{E}\{J_a^n\} = \exp\left\{\frac{\sigma_1^2 a^2 n^2}{2}\right\} K_{a,n},$$

$$K_{a,n} := \int_{[0,1]^n} \exp\left\{-\frac{a^2 n}{2} \sum_{i=1}^n q(t_i) - \frac{a^2}{2} \sum_{1 \leq i < j \leq n} \mathbb{E}\{(X(t_i) - X(t_j))^2\}\right\} dt.$$

Write $u := a^2 n$. If $\alpha \geq 1$ then

$$K_{a,n} \leq \left(\int_0^1 e^{-uq(t)/2} dt\right)^n \leq \left(\frac{C}{u}\right)^n.$$

If $\alpha < 1$ put

$$\Gamma(s, t) := q(s) + q(t) + \mathbb{E}\{(X(s) - X(t))^2\}, \quad B_u(x) := \int_0^1 e^{-u\Gamma(t,x)/4} dt.$$

Choose $U = [1 - \varepsilon_0, 1]$ with $0 < \varepsilon_0 < \varepsilon$. When $x \notin U$ the lower bound on $q(x)$ gives $B_u(x) \leq e^{-c\varepsilon_0 u/4}$. When $x \in U$ the integral over U^c is exponentially small and the increment lower bound gives

$$\int_U e^{-u\Gamma(t,x)/4} dt \leq \int_U e^{-c_d u|t-x|^\alpha/4} dt \leq C u^{-1/\alpha}.$$

Consequently $\sup_{x \in [0,1]} B_u(x) \leq C u^{-1/\alpha}$ for $u \geq 1$. The exponent in $K_{a,n}$ equals $-a^2 \sum_{i,j=1}^n \Gamma(t_i, t_j)/4$. For $D_j := \sum_{i=1}^n \Gamma(t_i, t_j)$, convexity gives

$$\exp\left\{-\frac{a^2}{4} \sum_{i,j=1}^n \Gamma(t_i, t_j)\right\} = \exp\left\{\frac{1}{n} \sum_{j=1}^n -\frac{u}{4} D_j\right\} \leq \frac{1}{n} \sum_{j=1}^n e^{-u D_j/4}.$$

Using $\Gamma(x, x) = 2q(x)$ and symmetry, we obtain

$$K_{a,n} \leq \int_0^1 e^{-uq(x)/2} B_u(x)^{n-1} dx \leq \left(C u^{-1/\alpha}\right)^{n-1}.$$

Thus uniformly for $a \geq 1$ and $n \in \mathbb{N}$

$$(71) \quad \log \mathbb{E}\{J_a^n\} \leq \frac{\sigma_1^2 a^2 n^2}{2} - \frac{n}{\gamma} \log(a^2 n) + O(n + \log(a^2 n)).$$

For $\lambda \geq 2$ let $n := \lfloor \lambda \rfloor$ and $p := \lambda/n \in [1, 2)$. Jensen's inequality on $[0, 1]$ gives $J_a^p \leq J_{pa}$ and hence $\mathbb{E}\{J_a^\lambda\} \leq \mathbb{E}\{J_{pa}^n\}$. Since

$$(pa)^2 n^2 = a^2 \lambda^2, \quad n \log\{(pa)^2 n\} = \lambda \log(a^2 \lambda) + O(1 + \log(a^2 \lambda))$$

(71) proves (70) with an implied constant independent of a and λ . □

A.2. Gaussian regression and short-interval estimates.

Lemma A.4. *Let $b > M_m(A)$, choose a minimiser h in (3), and write*

$$\rho := \rho_{A,m}(b), \quad f(t) := \frac{e^{m(t)+h(t)}}{b}, \quad V := V_A(f), \quad g := \frac{h}{\rho} = \frac{K_A f}{\sqrt{V}}.$$

Setting

$$(72) \quad Z := \frac{1}{\sqrt{V}} \int_A X(t) f(t) dt, \quad W(t) := X(t) - g(t)Z,$$

we have that Z is independent of W and

$$(73) \quad Z \sim N(0, 1), \quad \int_A W(t) f(t) dt = 0 \quad a.s.$$

Assume in addition that

$$(74) \quad g \geq 0 \quad \text{on } A, \quad \int_A \mathbf{1}_{\{g=0\}}(t) f(t) dt = 0.$$

Then for every $w \in C(A)$ there is a unique $\varkappa(w) \in \mathbb{R}$ such that

$$(75) \quad \int_A e^{\varkappa(w)g(t)+w(t)} f(t) dt = 1.$$

The map $\varkappa : C(A) \rightarrow \mathbb{R}$ is continuous in the uniform norm and

$$(76) \quad \varkappa(w) \leq -\frac{\int_A w(t)f(t) dt}{\sqrt{V}}, \quad \varkappa(W) \leq 0 \quad \text{a.s.}$$

Moreover, (5) is valid and in particular $\mathbb{P}\{I_{X,m}(A) = b\} = 0$, so (5) is unchanged if $>$ is replaced by $\geq$.

Remark A.5. If $A \subset [0, T]$ is a sufficiently small compact interval of positive length containing t_0 with $v(t_0) > 0$, then (5) holds for every $b > M_m(A)$. Indeed, continuity of R and $R(t_0, t_0) = v(t_0) > 0$ allow us to choose A such that

$$R(s, t) \geq \frac{v(t_0)}{2} > 0, \quad s, t \in A.$$

With the notation of Lemma A.4, since $\int_A f(t) dt = 1$, we obtain

$$g(t) = \frac{\int_A R(t, s)f(s) ds}{\sqrt{V}} \geq \frac{v(t_0)}{2\sqrt{V}} > 0, \quad t \in A$$

which verifies (74).

Proof of Lemma A.4. The integral defining Z is Gaussian, and (59) gives

$$\text{Var}(Z) = 1, \quad \text{Cov}(X(t), Z) = g(t), \quad \int_A g(t)f(t) dt = \sqrt{V}.$$

Standard Gaussian regression therefore gives $Z \perp W$, while the definitions give $\int_A W(t)f(t) dt = 0$ almost surely. This proves (73).

For $w \in C(A)$, set

$$F_w(x) := \int_A e^{xg(t)+w(t)} f(t) dt, \quad x \in \mathbb{R}.$$

Under (74), the function F_w is continuous and strictly increasing. Indeed, $g \geq 0$ everywhere and $g > 0$ for $f(t) dt$ -almost every t . For $x \leq 0$, the integrand is bounded by $e^{\|w\|_\infty} f(t)$, so dominated convergence gives

$$F_w(x) \longrightarrow 0 \quad \text{as } x \rightarrow -\infty.$$

Moreover, because $\{g > 0\} = \bigcup_{n \geq 1} \{g \geq 1/n\}$ has full $f(t) dt$ -measure, there is an $\varepsilon > 0$ for which

$$\int_{\{g \geq \varepsilon\}} f(t) dt > 0.$$

It follows that

$$F_w(x) \geq e^{x\varepsilon - \|w\|_\infty} \int_{\{g \geq \varepsilon\}} f(t) dt \longrightarrow \infty \quad \text{as } x \rightarrow \infty.$$

Thus there is a unique $\varkappa(w) \in \mathbb{R}$ satisfying (75).

For fixed $c \in \mathbb{R}$, the map $w \mapsto F_w(c)$ is continuous in the uniform norm. Strict monotonicity gives

$$\{w : \varkappa(w) < c\} = \{w : F_w(c) > 1\}, \quad \{w : \varkappa(w) > c\} = \{w : F_w(c) < 1\}.$$

Both sets are open in $C(A)$; hence $\varkappa$ is continuous.

Applying Jensen's inequality to (75) yields

$$\begin{aligned} 0 &= \log F_w(\varkappa(w)) \geq \varkappa(w) \int_A g(t)f(t) dt + \int_A w(t)f(t) dt \\ &= \varkappa(w)\sqrt{V} + \int_A w(t)f(t) dt. \end{aligned}$$

This proves the first inequality in (76); the second follows by taking $w = W$ and using (73). Since $h = \rho g$ and $e^{m+\rho g} = bf$, we have

$$\begin{aligned} I_{X,m}(A) &= \int_A e^{m(t)+X(t)} dt \\ &= b \int_A e^{X(t)-\rho g(t)} f(t) dt \\ &= b \int_A e^{(Z-\rho)g(t)+W(t)} f(t) dt = bF_W(Z - \rho). \end{aligned}$$

Since F_W is strictly increasing,

$$I_{X,m}(A) > b \iff Z > \rho + \varkappa(W).$$

Conditioning on W and using $Z \perp W$ and $Z \sim N(0, 1)$ proves (5). Similarly,

$$\{I_{X,m}(A) = b\} = \{Z = \rho + \varkappa(W)\},$$

which has probability zero after conditioning on W . □

Lemma A.6. *For each b , let*

$$Q_b = [u_b, u_b + r_b], \quad r_b > 0,$$

and let $X_b(t), t \in Q_b$ be a centred Gaussian process with continuous sample paths and covariance R_b . Suppose that for some positive numbers τ_b , fixed constants $c_0, C_0, C_1 > 0$ and a fixed exponent $\alpha \in (0, 2]$

$$(77) \quad c_0 \tau_b^2 \leq R_b(s, t), \quad R_b(t, t) \leq C_0 \tau_b^2, \quad s, t \in Q_b,$$

and

$$\mathbb{E}\{(X_b(s) - X_b(t))^2\} \leq C_1 \tau_b^2 |s - t|^\alpha, \quad s, t \in Q_b.$$

Let $m_b \in C(Q_b)$, and put

$$\chi_b := r_b^\alpha, \quad M_b := \int_{Q_b} e^{m_b(t)} dt.$$

For $c_b > M_b$, define

$$\ell_b := \log \frac{c_b}{M_b}, \quad \rho_b := \rho_{Q_b, m_b}(c_b),$$

where ρ_b is computed for the process X_b . Then

$$(78) \quad \frac{\ell_b}{\tau_b \sqrt{C_0}} \leq \rho_b \leq \frac{\ell_b}{\tau_b \sqrt{c_0}}.$$

If

$$(79) \quad (\tau_b + \ell_b)\chi_b \longrightarrow 0, \quad b \rightarrow \infty,$$

then, for all sufficiently large b

$$(80) \quad 1 \leq \frac{\mathbb{P}\left\{\int_{Q_b} e^{X_b(t)+m_b(t)} dt > c_b\right\}}{\Psi(\rho_b)} \leq 1 + C(\tau_b + \ell_b)\chi_b,$$

where C is independent of b .

Proof. All constants below depend only on c_0, C_0, C_1, α . For every admissible density f ,

$$\ell_b + D_{Q_b, m_b}(f) \geq \ell_b, \quad V_{Q_b}(f) \leq C_0 \tau_b^2.$$

The variational formula gives the lower level bound. For the reverse bound setting

$$q_b(t) := \frac{e^{m_b(t)}}{M_b}$$

we have $D_{Q_b, m_b}(q_b) = 0$ and $V_{Q_b}(q_b) \geq c_0 \tau_b^2$ proving (78).

Choose an RKHS minimiser and write

$$f_b(t) := \frac{e^{m_b(t) + h_b(t)}}{c_b}, \quad V_b := V_{Q_b}(f_b), \quad k_b(t) := \int_{Q_b} R_b(t, s) f_b(s) ds, \quad g_b := \frac{k_b}{\sqrt{V_b}}.$$

By Lemma A.1, $h_b = \rho_b g_b$. The covariance bounds imply

$$c_0 \tau_b^2 \leq V_b \leq C_0 \tau_b^2, \quad c_0 \tau_b^2 \leq k_b(t) \leq C_0 \tau_b^2,$$

and hence, for fixed $0 < \underline{c} \leq \bar{c} < \infty$

$$(81) \quad \tau_b \underline{c} \leq g_b(t) \leq \tau_b \bar{c}, \quad t \in Q_b.$$

In particular, the sign condition (74) is satisfied.

If Z_b, W_b are the regression variables from Lemma A.4, then

$$\int_{Q_b} W_b(t) f_b(t) dt = 0, \quad \mathbb{P} \left\{ \int_{Q_b} e^{X_b(t) + m_b(t)} dt > c_b \right\} = \mathbb{E} \{ \Psi(\rho_b - x_b) \},$$

where

$$x_b := -\mathcal{K}_b(W_b) \geq 0.$$

We use the following deterministic estimate. If μ is a probability measure, w is bounded and measurable, and

$$g \geq m_0 > 0, \quad \int w d\mu = 0, \quad \int e^{w - xg} d\mu = 1, \quad x \geq 0,$$

then

$$1 = \int e^{w - xg} d\mu \leq e^{-m_0 x} \int e^w d\mu.$$

Hoeffding's lemma,¹ applied under μ to the bounded mean-zero function w , therefore gives

$$(82) \quad x \leq \frac{1}{m_0} \log \int e^w d\mu \leq \frac{(\text{osc } w)^2}{8m_0}.$$

We apply this estimate pathwise with $\mu(dt) = f_b(t) dt$, $w = W_b$, $g = g_b$, and $m_0 = \tau_b \underline{c}$. Indeed, μ is a probability measure, almost every sample path of W_b is bounded on the compact interval Q_b , and $\int_{Q_b} W_b(t) f_b(t) dt = 0$. Together with (81), this gives

$$(83) \quad 0 \leq x_b \leq C \frac{\omega_b^2}{\tau_b}, \quad \omega_b := \text{osc}_{Q_b} W_b.$$

The regression identity gives

$$\mathbb{E} \{ (W_b(s) - W_b(t))^2 \} = \mathbb{E} \{ (X_b(s) - X_b(t))^2 \} - \{ g_b(s) - g_b(t) \}^2 \leq C_1 \tau_b^2 |s - t|^\alpha.$$

Define the anchored, rescaled process on $[0, 1]$ by

$$\widetilde{W}_b(u) := \frac{W_b(u_b + r_b u) - W_b(u_b)}{\tau_b r_b^{\alpha/2}}, \quad 0 \leq u \leq 1.$$

Its canonical metric $\widetilde{d}_b$ and pointwise variances satisfy

$$\widetilde{d}_b(u, v) \leq \sqrt{C_1} |u - v|^{\alpha/2}, \quad \sup_{0 \leq u \leq 1} \text{Var} \{ \widetilde{W}_b(u) \} \leq C_1.$$

¹W. Hoeffding, Probability inequalities for sums of bounded random variables, *Journal of the American Statistical Association* **58** (1963), 13–30, p. 22, Eq. (4.16); doi:10.1080/01621459.1963.10500830.

Consequently, $N([0, 1], \widetilde{d}_b, \varepsilon) \leq 1 + C\varepsilon^{-2/\alpha}$, uniformly in b and $\varepsilon > 0$. Dudley's bound gives a uniform bound for $\mathbb{E}\{\sup_u |\widetilde{W}_b(u)|\}$, and Borell–TIS, applied also to $-\widetilde{W}_b$, gives a uniform sub-Gaussian tail. Since

$$\omega_b = \tau_b r_b^{\alpha/2} \text{osc}_{[0,1]} \widetilde{W}_b,$$

there are constants $\eta, K > 0$, independent of b , such that

$$(84) \quad \mathbb{E} \left\{ \exp \left\{ \eta \frac{\omega_b^2}{\tau_b^2 \chi_b} \right\} \right\} \leq K.$$

The standard normal hazard rate $h_N(u) := \varphi(u)/\Psi(u)$ is increasing and, for $u \geq 0$,

$$h_N(u) \leq \frac{1}{2} \{u + \sqrt{u^2 + 4}\} \leq 1 + u.$$

Therefore, for $x \geq 0$, since $\rho_b > 0$,

$$\log \frac{\Psi(\rho_b - x)}{\Psi(\rho_b)} = \int_{\rho_b - x}^{\rho_b} h_N(u) du \leq x h_N(\rho_b) \leq (1 + \rho_b)x,$$

and hence

$$1 \leq \frac{\Psi(\rho_b - x)}{\Psi(\rho_b)} \leq e^{(1 + \rho_b)x}.$$

Combining this with (83) gives

$$\frac{\Psi(\rho_b - x_b)}{\Psi(\rho_b)} \leq \exp \left\{ \Lambda_b \frac{\omega_b^2}{\tau_b^2 \chi_b} \right\}, \quad \Lambda_b := C\tau_b(1 + \rho_b)\chi_b.$$

The level comparison (78) implies

$$\Lambda_b \leq C(\tau_b + \ell_b)\chi_b \longrightarrow 0, \quad b \rightarrow \infty.$$

For large b , convex interpolation between 1 and the exponential in (84) yields

$$\mathbb{E} \left\{ \exp \left\{ \Lambda_b \frac{\omega_b^2}{\tau_b^2 \chi_b} \right\} \right\} \leq 1 + C\Lambda_b.$$

Consequently we have

$$1 \leq \frac{\mathbb{P}\{\int_{Q_b} e^{X_b(t) + m_b(t)} dt > c_b\}}{\Psi(\rho_b)} \leq 1 + C\tau_b(1 + \rho_b)\chi_b \leq 1 + C'(\tau_b + \ell_b)\chi_b$$

establishing the proof. $\square$

A.3. Absolute continuity. The following elementary lemma is a specialization of the classical stratification method for Gaussian functionals; see [20, 21, 22]. We give the short argument because it makes the required nondegeneracy particularly transparent.

Lemma A.7. *Let $T > 0$ and let $X(t), t \in D := [0, T]^d$ be a real Gaussian random field with almost surely continuous sample paths where $d \geq 1$. If $\text{Var}(X(t_0)) > 0$ for some $t_0 \in D$, then*

$$I_D := \int_D e^{X(t)} dt$$

has a distribution that is absolutely continuous with respect to Lebesgue measure.

Proof. Setting next

$$\mu(t) := \mathbb{E}\{X(t)\}, \quad \sigma^2 := \text{Var}(X(t_0)) > 0, \quad Z := \frac{X(t_0) - \mu(t_0)}{\sigma}$$

we have that Z is a standard Gaussian random variable. Define

$$a(t) := \text{Cov}(X(t), Z), \quad Y(t) := X(t) - \mu(t) - a(t)Z.$$

The functions μ and a are continuous, and Y has continuous sample paths. Indeed, sample continuity and Gaussianity imply continuity of the mean and covariance functions. Moreover, we have

$$\text{Cov}(Y(t), Z) = 0, \quad t \in D,$$

so joint Gaussianity gives independence of Z from every finite evaluation vector of Y . Evaluations on a countable dense subset of D generate the Borel σ -field of the separable space $C(D)$. Hence Z and the $C(D)$ -valued random element Y are independent.

For $y \in C(D)$, define

$$\Phi_y(z) := \int_D \exp\{\mu(t) + y(t) + a(t)z\} dt, \quad z \in \mathbb{R}.$$

The function Φ_y is smooth, with

$$\Phi_y''(z) = \int_D a(t)^2 \exp\{\mu(t) + y(t) + a(t)z\} dt.$$

Since

$$a(t_0) = \text{Cov}(X(t_0), Z) = \sigma > 0$$

and a is continuous, there is a relative neighbourhood of t_0 in D on which $a \geq \sigma/2$. This neighbourhood has positive d -dimensional Lebesgue measure, including when t_0 is on the boundary of D . Consequently,

$$\Phi_y''(z) > 0, \quad z \in \mathbb{R}.$$

Hence Φ_y' is strictly increasing and has at most one zero.

Let $A \subset \mathbb{R}$ be a Lebesgue-null Borel set. Outside the possible single zero of Φ_y' , cover $\mathbb{R}$ by countably many compact intervals K on which $\inf_{z \in K} |\Phi_y'(z)| > 0$. The restriction of Φ_y to each K is strictly monotone with a Lipschitz inverse; the possible critical point contributes at most one point to the preimage. Consequently $\Phi_y^{-1}(A)$ is Lebesgue-null and hence also null for the standard Gaussian law γ . The map $(y, z) \mapsto \Phi_y(z)$ is jointly continuous. Independence of Y and Z , followed by Fubini's theorem, now gives

$$\mathbb{P}\{I_D \in A\} = \int_{C(D)} \gamma(\Phi_y^{-1}(A)) \mathbb{P}_Y(dy) = 0.$$

Thus the law of I_D is absolutely continuous. □

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